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Lab-08

CircleSizeSlider

a. What function runs to change the size of the circle?

The function that responds to changes in the slider is the **event handler attached to the JSlider**, typically a `ChangeListener`. When the slider value changes, the listener updates the circle's radius variable.

b. When and how often is the circle redrawn?

The circle is redrawn **every time the slider value changes**. Each adjustment triggers a repaint call, which causes the drawing method to be executed again.

Random Programs

a. Where are these programs getting random data from?

The programs retrieve random data from Java's built-in random number generation utilities.

b. Is the data truly random or pseudo-random?

The data is **pseudo-random**, because it is generated algorithmically. Although it appears random, it is ultimately determined by an initial seed value.

LogisticMapGenerator

a. What range of growth values lead to extinction?

Growth values **below 1.0** lead to extinction. In this range, the population steadily decreases until it reaches zero.

b. What range of growth values lead to convergence to a constant population?

Growth values roughly between **1.0 and 3.0** lead to convergence, where the population stabilizes at a single fixed value over time.

c. What range of growth values lead to oscillating or chaotic populations?

Growth values **above 3.0**, especially beyond **3.57**, lead to oscillations and chaotic behavior, where the population fluctuates unpredictably.

Visualizing

a. What did changing parameters help you understand?

Making small changes such as graph color, spacing, or animation timing helped clarify how data is read line-by-line from CSV files and transformed into visual output. Adjusting timers also made population trends easier to observe.

GUI Interface (RabbitPopulationGraph)

b. What events trigger the graph to be redrawn?

The graph redraws when **input values change**, such as growth rate or initial population. GUI events like slider movement or button presses trigger recalculations and repainting.

c. Did any conclusions change after using the GUI?

The GUI confirmed earlier conclusions but made chaotic behavior more obvious. Some growth values that seemed stable numerically showed subtle oscillations when visualized.

Bifurcation Diagram

The diagram shows how long-term population values split as growth rates increase. The branching patterns represent transitions from stable populations to oscillations and chaos. Empty regions indicate population values that cannot occur at certain growth rates.

Agentic Simulation

a. What determines the rabbit population limit?

The rabbit population is limited by **resource availability**, primarily the number of carrots available for consumption.

b. How is the carrot population maintained?

Carrots are replenished according to growth rules built into the simulation, simulating natural regrowth over time.

c. How close is the simulation to the logistic map predictions?

The simulation closely follows logistic map trends but includes natural variation due to individual agent interactions. This makes it slightly less precise.